Artificial Intelligence Techniques - 10211CS211

Assignment

Name: S. Sarika

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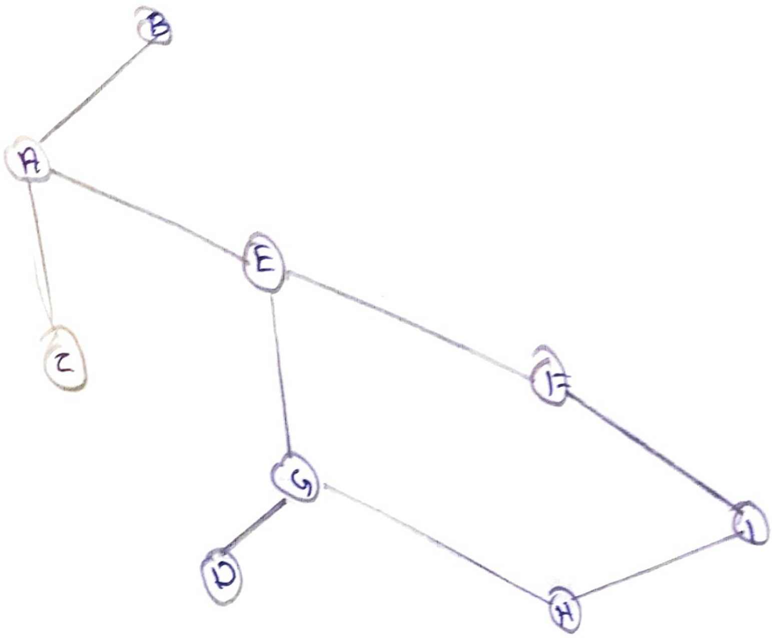
Slot : S_2L_{13}

Define BFS and find the shortest path from
to I.



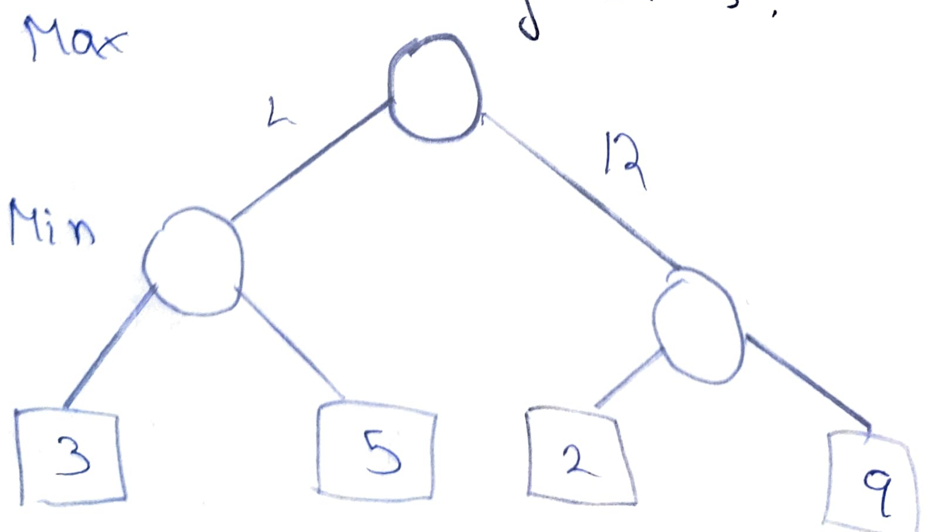
Step	Current Node n	Neighbors m of n	D[n] + cost(n, m)	New D[m]	Unvisited nodes
0	-	-	-	D[A] = 0	A: 0, F
1	A (D = 0)	B	0+75 = 75	D[B] = 75	C: 118, D: 75.
		C	0+118 = 118	D[C] = 118	B: 75, C: 118
		E	0+140 = 140	D[E] = 140	B: 75, C: 118, E: 140
2	B (D = 75)	Unvisited nodes	-	-	C: 118, E: 140
3	C (D = 118)	D	118 + 111 = 229	D[D] = 229	E: 140, D: 229
4	E (D = 140)	F	140 + 99 = 239	D[F] = 239	D: 229, F: 239

5	G(D=220)	H	220+97 = 317	D[H] = 317	D: 229 F: 2.. H: 317
6	D(D= 229)	Not visitied	-	-	F: 239 H:
7	F(D= 239)	I	239+ 211=450	D[I] = 450	H: 317 I: 4..
8	H(D=317)	I	317+101 =418	D[I] =418	I: 418
9	I(D: 418	Goal reached			

Shortest path: A → E → G → H → I

140+80+95+101 = 418

2 Apply the Minimax Algorithm to the following game Scenario and determine the optimal move for MAX player. Assu.. it's two-player, turn based Zero-Sum with alternating turns.

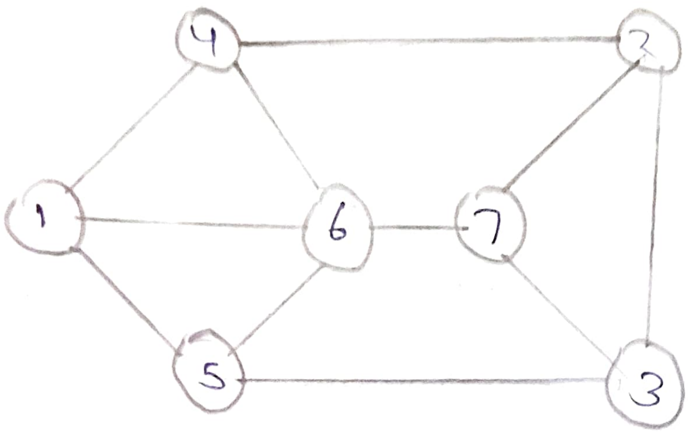


MIN NODE	Children Value	Min choose
L (Left Subtree)	{3, 5}	min(3, 5) = 3
R (Right Subtree)	{2, 9}	min(2, 9) = 2

Level	Player	Nodes to evaluate	Calculation	Node value
3	Terminal	Leaf Node		3, 5, 2, 9
2	MIN	Left child	min(3, 5)	3
	MIN	Right child	min(2, 9)	2
1	MAX	Root	max(Value of L, Value of R	max(3, 2) = 3.

game ends at final score 3.

A Communication network Const of Seven routers. Each router must operate on a distinct frequency from any directly connected router to avoid single interface. A Constraint is place that Only three frequencies are available. Blue, Green, Red.



Router	frequency	Neighbors
1	G	4(R), 5(R), 6(B)
2	B	4(R), 7(R), 3(G)
3	G	2(B), 7(R), 5(R)
4	R	1(G), 6(B), 2(B)
5	R	1(G), 6(B), 3(G)
6	B	1(G), 4(R), 5(R), 7(R)
7	R	6(R), 2(B), 3(G)

Minimum No. of frequency : 3.

4. Solve the following sentence in Conceptual graph.
- "A message is between sender and receiver
- "John knows that sun rises in east"

1. [Message] → (Betw) → [SENDER]
↑
(AND) → [RECEIVER]

2. [PERSON: JOHN] ← (AGNT) ← [KNOW] → (OBJ) → [PROPOSITION: [SUN] ← (AGNT) ← [RISE] → (LOC) → [EAST]].

5. Proove by resolution that Ravi likes Peanuts using resolution.

a. Ravi Likes all kinds of food.

b. Apples and chicken are food.

c. Anything anyone eats is not killed is food.

d. Ajay eats peanuts and is still alive.

Rita eats everything that Ajay eats

$L(x, y)$: x likes y

$F(x)$: x is food

$E(x, y)$: x eats y

$K(x)$: x is killed.

Rani (r), Ajay (a), Food (F), Apple (ap), Chicken (c).

Sentence	Predicate form	CNF
Rani likes all kind food	$\forall x (F(x) \rightarrow L(r, x))$	$\neg F(x) \vee L(r,$ $(c,)$
Apples and Chicken are food	$F(ap) \wedge F(c)$	$F(ap)$ $F(c)$
Anything is food not killed	$E(a, p) \wedge \neg K(a)$	$E(a, p)$
Ajay eats peanut still alive	$E(a, p) \wedge \neg K(a)$	$E(a, p)$
Rita eats everything that Ajay eats	$\forall x (E(a, x) \rightarrow E(r, x))$	$\neg E(a, x) \vee E(r, x)$